MANAGERIAL ACCOUNTING

Comprehensive Exam Study Guide

International MBA Program

IE University – IE Business School

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Sessions Covered:	1 – 11 (of 15)
Term:	Core II, SEP-2025 S-4
Area:	Accounting and Management Control

Key Topics: Cost Classification • Job & Process Costing • Cost Allocation Methods • Joint Costs • Capacity Analysis • ABC & TDABC • Budgeting • Variable vs. Absorption Costing • CVP Analysis

Case Studies: Ecoservicios S.L. • Terminus Hotel (A & B) • Enred@me

Tools & Frameworks: Direct/Step-Down/Reciprocal Allocation • Cost Hierarchies • Master Budgets • Flexible Budgets • Break-Even Analysis • Operating Leverage

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1 Introduction to Managerial Accounting (Session 1)

1.1 What Is Managerial Accounting?

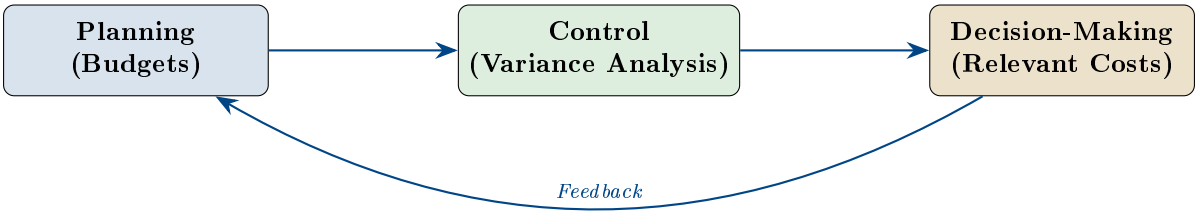
Managerial (or management) accounting is the branch of accounting that focuses on providing information to **internal users**—managers at all levels—for the purposes of planning, controlling, and decision-making within an organization. Unlike financial accounting, which produces standardized reports for external stakeholders (investors, creditors, regulators), managerial accounting is flexible, forward-looking, and tailored to the specific needs of the organization.

Dimension	Financial Accounting	Managerial Accounting
Purpose	External reporting	Internal decision-making
Users	Investors, creditors, regulators	Managers, employees
Focus	Past performance	Future-oriented
Rules	GAAP/IFRS required	No mandatory standards
Output	Financial statements	Reports, analyses, budgets
Behavior	Compliance-driven	Decision-support driven

1.2 The Three Key Management Activities

Managerial accounting supports three critical activities:

1. **Planning:** Setting objectives and outlining courses of action. Budgets are the primary tool.
2. **Control:** Monitoring actual performance against plans and taking corrective action. Variance analysis is the primary tool.
3. **Decision-Making:** Choosing among alternatives using relevant cost and revenue information.



1.3 Three Guiding Principles

When making cost-related decisions, managers rely on three guiding principles:

1. **Cost–Benefit Principle:** The benefits of an action or information system should exceed its costs. Spending resources on a more refined costing system is only justified if the improved information leads to better decisions.
2. **Behavioral and Technical Implications:** Cost systems influence behavior. Managers respond to incentives embedded in cost measurement. Both the technical accuracy and the behavioral consequences of a costing system matter.

3. **Different Costs for Different Purposes:** There is no single "correct" cost. The appropriate cost depends on the purpose—pricing, performance evaluation, financial reporting, or strategic decision-making all may require different cost figures.

There is **no single "scientific" cost**. As illustrated in the Ecoservicios case, the same project generated four different cost estimates depending on the assumptions and purposes. Always ask: *"What is the purpose of this cost calculation?"*

1.4 Framework for the Course

The course is structured around three pillars:

Pillar 1	**Pillar 2**	**Pillar 3**
Calculate cost of products, services (Sessions 2–8)	Planning, control & performance evaluation (Sessions 9–10)	Relevant information for decisions (Sessions 11–14)

1.5 ESG and Management Accounting

Environmental, Social, and Governance (ESG) factors are increasingly integrated into managerial accounting. Key considerations include:

- **Environmental:** Conservation costs, carbon footprint measurement, waste management costs
- **Social:** Stakeholder relationship costs, employee well-being, community impact
- **Governance:** Internal controls, ethical compliance, transparency in reporting

Cost accounting provides valuable information for financial and non-financial decisions. Cost calculations always depend on purpose and should never be considered on a standalone basis. As ESG becomes more prominent, management accounting has a vital role to play in measuring and controlling these factors.

2 Cost Terms and Classifications (Session 2)

2.1 Basic Cost Terminology

Fundamental Definitions

Cost: A sacrificed resource (monetary measure) to achieve a specific objective.

Cost Object: Anything of interest for which a cost is desired—a product, service, department, project, activity, or customer.

2.2 Cost Behavior Classification: Fixed vs. Variable

2.2.1 Variable Costs

Variable costs change **in total** in direct proportion to changes in the level of activity or volume of output within the relevant range.

Variable Cost Formulas

$$\text{Total Variable Cost} = \text{Variable Cost per Unit} \times \text{Quantity (Q)}$$

The **unit variable cost remains constant** regardless of volume.

2.2.2 Fixed Costs

Fixed costs remain **unchanged in total** regardless of changes in activity level within the relevant range.

Fixed Cost Behavior

$$\text{Total Fixed Cost} = \text{Constant (within relevant range)}$$

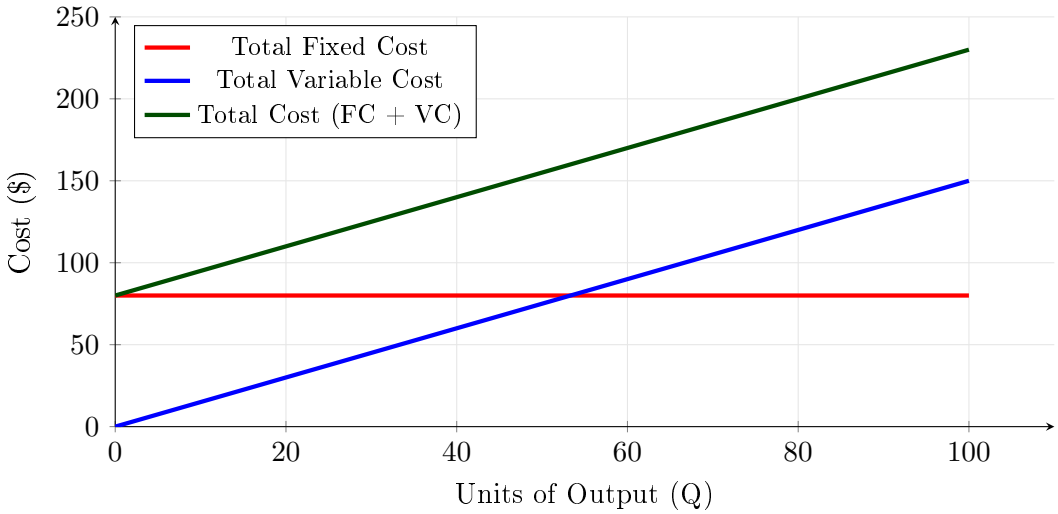
$$\text{Fixed Cost per Unit} = \frac{\text{Total Fixed Cost}}{Q}$$

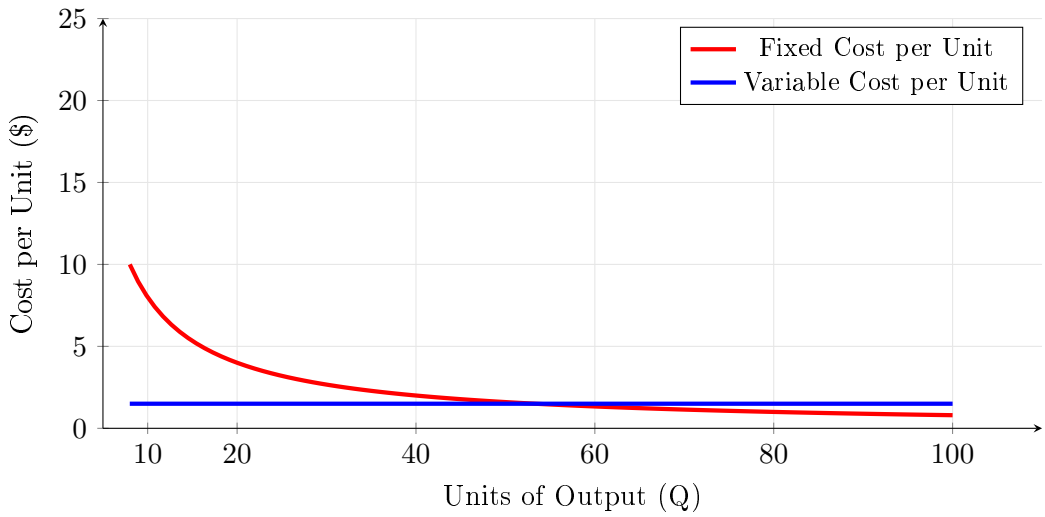
The unit fixed cost **decreases** as volume increases (inverse relationship).

2.2.3 Mixed Costs

Some costs contain both fixed and variable components:

- **Step costs:** Fixed within specified activity levels, but jump by a constant amount at critical thresholds.
- **Semi-variable costs:** Include both a fixed and variable component, expressed as $Y = a + bX$, where a is the fixed component and b is the variable rate per unit of activity.





Unit costs should be used cautiously. Since unit fixed costs change with different levels of output, it may be more prudent to base decisions on a total dollar basis rather than per-unit figures. Misusing unit costs can lead to incorrect pricing and decision-making.

2.2.4 The Relevant Range

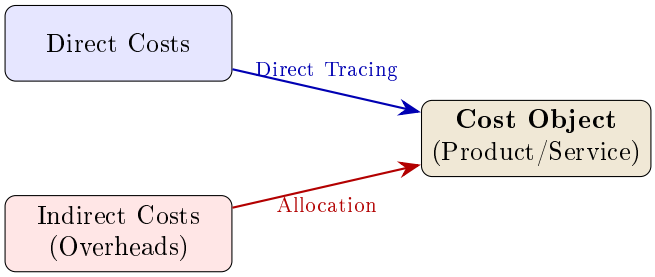
The **relevant range** is the band of normal activity level or volume within which a specific cost–behavior relationship holds. Outside this range, cost behavior may change (e.g., fixed costs may increase in steps).

2.3 Cost Object Classification: Direct vs. Indirect

Direct vs. Indirect Costs

Direct Costs: Costs that can be *conveniently and economically traced* (tracked) to a specific cost object. Examples: raw materials used in a product, labor of a worker assembling a specific item.

Indirect Costs (Overheads): Costs that *cannot* be conveniently or economically traced to a specific cost object. Instead, they are *allocated* using a rational and systematic method. Examples: factory rent, supervisory salaries, utilities.



A **cost driver** is whatever generates a cost in a given time span (cause-and-effect relationship). Traditionally, the volume of production was used; modern systems use activity-based drivers.

2.4 Combining Classifications

	Direct Costs	**Indirect Costs**
Variable Costs	Direct materials, Direct labor	Energy for machines used on multiple products
Fixed Costs	Depreciation of equipment for a single product	Rent of building used for several products

2.5 General Income Statement Classification: Inventoriable vs. Period

Income Statement Structure

Revenues (Sales)
− Cost of Goods Sold ← *Inventoriable (Product) Costs*
= Gross Margin
− Operating Expenses ← *Period Costs*
= Operating Income

Inventoriable (Product) Costs: Costs tied to the acquisition or production of goods. They are capitalized as inventory on the balance sheet until the goods are sold, at which point they become Cost of Goods Sold (COGS). For manufacturers: Direct Materials + Direct Labor + Manufacturing Overhead.

Period Costs: Costs expensed in the period they are incurred. They are *not* attached to inventory. Examples: marketing, distribution, administration, R&D.

2.6 Manufacturing Cost Classification

For manufacturing organizations, three inventory accounts are maintained:

1. **Raw Materials Inventory:** Cost of materials purchased but not yet used.
2. **Work-in-Process (WIP) Inventory:** Cost of partially completed goods.
3. **Finished Goods Inventory:** Cost of completed goods awaiting sale.

Cost of Goods Manufactured (COGM)

$$\text{Beginning DM Inventory} + \text{Purchases of DM} - \text{Ending DM Inventory} = \text{DM Used}$$
$$\text{DM Used} + \text{Direct Manufacturing Labor} + \text{Manufacturing Overhead} = \text{Total Mfg. Cost}$$
$$\text{Beginning WIP} + \text{Total Mfg. Cost} - \text{Ending WIP} = \textbf{COGM}$$

Cost of Goods Sold (COGS)

$$\text{COGS} = \text{Beginning FG Inventory} + \text{COGM} - \text{Ending FG Inventory}$$

Incorrectly classifying period costs as product costs (or vice versa) can manipulate reported income. If warehousing costs (a period expense) are reclassified as product costs, some costs shift to unsold inventory on the balance sheet, inflating current-period operating income.

Master the cost classifications: (1) Behavior: Fixed vs. Variable, (2) Cost Object: Direct vs. Indirect, (3) General: Inventoriable vs. Period. Understand how misclassification affects financial reporting and managerial decisions. Always consider whether you are looking at total costs or unit costs.

3 Building a Costing System – Job Costing (Session 3)

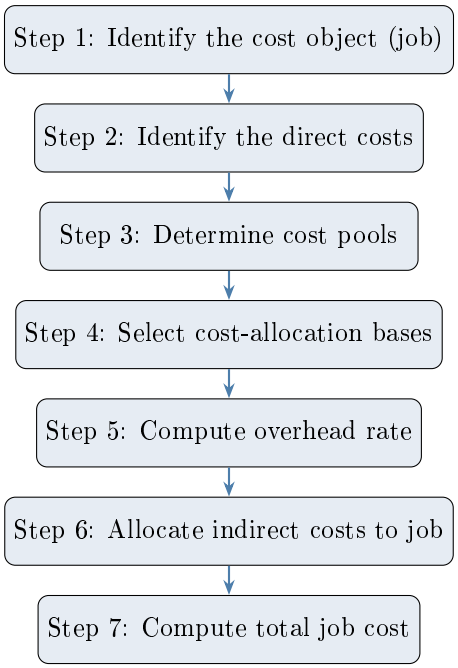
3.1 The Five Building Blocks of a Costing System

1. **Cost Object:** Anything for which we desire a cost measurement.
2. **Direct Costs:** Costs that can be traced in an economically feasible way.
3. **Indirect Costs:** Costs that cannot be traced economically.
4. **Cost Pool:** A grouping of individual indirect cost items.
5. **Cost-Allocation Base:** The systematic way to link indirect costs to cost objects (cause and effect).

3.2 Job Costing vs. Process Costing

Job Costing	Process Costing
Cost object is a distinct product or service (the "job") Unique, customized outputs Examples: advertising campaigns, building construction, drug development	Cost object is a mass-produced, indistinct product Homogeneous, continuous outputs Examples: oil refining, mining, juice production

3.3 The Seven Steps of Job Costing



3.4 Normal Costing vs. Actual Costing

Overhead Rate Calculation

Budgeted (Normal) Overhead Rate:

$$\text{Budgeted OH Rate} = \frac{\text{Budgeted Manufacturing Overhead Costs}}{\text{Budgeted Total Quantity of Cost-Allocation Base}}$$

Actual Overhead Rate:

$$\text{Actual OH Rate} = \frac{\text{Actual Manufacturing Overhead Costs}}{\text{Actual Total Quantity of Cost-Allocation Base}}$$

Cost Component	Normal Costing	Actual Costing
Direct Materials	Actual rates × Actual qty	Actual rates × Actual qty
Direct Labor	Actual rates × Actual hrs	Actual rates × Actual hrs
Manufacturing OH	**Budgeted rate** × Actual base	Actual rate × Actual base

Normal costing is preferred because: (1) Actual overhead rates are only known at year-end, making timely job costing impossible. (2) Actual rates fluctuate due to seasonal variations, creating inconsistent product costs. (3) Budgeted rates provide stability for pricing and planning.

3.5 Over- and Under-Allocation of Overhead

Since normal costing uses budgeted rates, the total overhead *allocated* to jobs rarely equals the *actual* overhead incurred.

Over/Under-Allocation

$$\text{Over/Under-Allocated OH} = \text{Actual OH Incurred} - \text{OH Allocated to Jobs}$$

- If Actual > Allocated ⇒ **Under-allocated** (unfavorable)
- If Actual < Allocated ⇒ **Over-allocated** (favorable)

End-of-year adjustment: typically written off to Cost of Goods Sold.

Caldwell Toys (Exercise 4-23)

Given: Budgeted MOH = \$5,100,000; Budgeted DLH = 150,000; Actual MOH = \$5,355,000; Actual DLH = 153,000.

Budgeted Rate: $\frac{\$5{,}100{,}000}{150{,}000} = \$34/\text{DLH}$

Actual Rate: $\frac{\$5{,}355{,}000}{153{,}000} = \$35/\text{DLH}$

Steel Wheels (Normal Costing): $\$78{,}290 + \$25{,}445 + (840 \times \$34) = \$78{,}290 + \$25{,}445 + \$28{,}560 = \$132{,}295$

Magic Wheels (Normal Costing): \$94,650+\$32,752+(960×\$34) = \$94,650+\$32,752+\$32,640 = \$160,042

Understand the five building blocks and seven steps of job costing. Normal costing uses budgeted overhead rates for timeliness and consistency. The consequence is potential over- or under-allocation of overhead that requires end-of-year adjustment. Cross-subsidization between products (undercosting/overcosting) is a concern.

4 Allocation of Period and Joint Costs (Sessions 4–5)

4.1 Why Allocate Costs?

1. To provide information for **economic decisions**
2. To **motivate** managers and employees
3. To **justify costs** or compute reimbursement amounts
4. To measure income and assets for **tax reporting**

4.2 Criteria for Cost Allocation Decisions

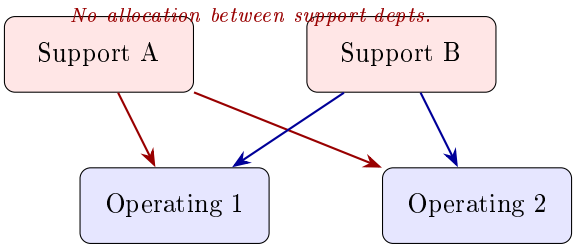
Criterion	Description
Cause and Effect	"You pay for what you used." Most credible to managers. Integral to ABC.
Benefits Received	"You pay for what you gain." Costs charged in proportion to benefits.
Fairness (Equity)	Allocation is a "reasonable" or "fair" basis for establishing selling price.
Ability to Bear	Costs allocated in proportion to financial strength or profitability.

4.3 Allocating Support Department Costs

Operating (production) departments directly add value to products or services. **Support (service) departments** provide services to operating departments and other support departments.

4.3.1 Method 1: Direct Method

Allocates support-department costs **only to operating departments**. Ignores any services provided between support departments.



Direct Method Allocation

When allocating Support Department A's costs to operating departments, the usage percentages of other support departments are **ignored**. Recalculate percentages using only operating departments:

$$\text{Allocation to Op}_i = \text{Support Cost} \times \frac{\text{Op}_i\text{'s usage}}{\text{Total usage by all operating depts.}}$$

4.3.2 Method 2: Step-Down (Sequential) Method

Allocates support-department costs in a **sequential manner** that partially recognizes mutual services. Once a support department's costs have been allocated, it receives no further allocations.

The order in which support departments are allocated matters and affects the final results. Typically, the department providing the most service to other support departments is allocated first.

4.3.3 Method 3: Reciprocal Method

Fully recognizes mutual services among all support departments using simultaneous linear equations.

Reciprocal Method – Linear Equations

Three steps:

1. Express support department costs as linear equations incorporating reciprocal relationships.
2. Solve the system of equations to obtain complete reciprocated costs.
3. Allocate complete reciprocated costs to all departments using original usage percentages.

Example: If AS and IS are support departments:

$$\begin{aligned} \text{AS}_{\text{total}} &= \text{AS}_{\text{budget}} + 0.10 \times \text{IS}_{\text{total}} \\ \text{IS}_{\text{total}} &= \text{IS}_{\text{budget}} + 0.25 \times \text{AS}_{\text{total}} \end{aligned}$$

Solve simultaneously, then allocate using original percentages.

Phoenix Partners (E16-19/20)

Given: AS budget = \$600,000; IS budget = \$2,400,000.
AS provides: 25% to IS, 40% to GOVT, 35% to CORP.
IS provides: 10% to AS, 30% to GOVT, 60% to CORP.

Direct Method (AS to operating only): AS to GOVT: $600{,}000 \times \frac{40}{40+35} = 600{,}000 \times \frac{40}{75} = \$320{,}000$

AS to CORP: $600{,}000 \times \frac{35}{75} = \$280{,}000$

Reciprocal Method:

$$\begin{aligned} \text{AS} &= 600{,}000 + 0.10 \times \text{IS} \\ \text{IS} &= 2{,}400{,}000 + 0.25 \times \text{AS} \end{aligned}$$

Substituting: $\text{AS} = 600{,}000 + 0.10(2{,}400{,}000 + 0.25\text{AS})$
$\text{AS} = 600{,}000 + 240{,}000 + 0.025\text{AS}$
$0.975\text{AS} = 840{,}000 \Rightarrow \text{AS} = \$861{,}538$
$\text{IS} = 2{,}400{,}000 + 0.25(861{,}538) = \$2{,}615{,}385$

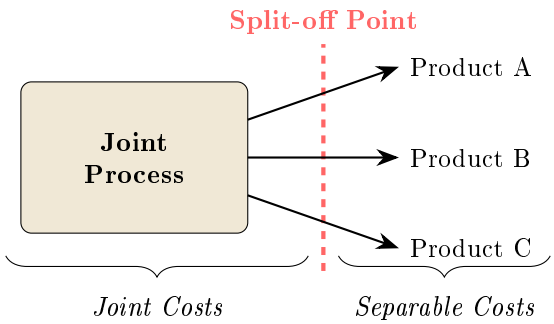
Then allocate using original percentages.

Direct	Step-Down	Reciprocal
Simplest to compute	Partially recognizes interdependence	Most precise
Ignores support-to-support services	Order matters	Fully accounts for mutual services
Most widely used	Moderate complexity	Requires solving linear equations

4.4 Joint Cost Allocation

4.4.1 Key Definitions

- **Joint costs:** Costs of a production process that yields multiple products simultaneously.
- **Split-off point:** The moment when products become distinguishable.
- **Separable costs:** Costs incurred *after* the split-off point, assignable to specific products.
- **Joint products:** Outputs with relatively *high* sales values.
- **Byproducts:** Outputs with relatively *low* sales values.



4.4.2 Joint Cost vs. Overheads

Joint Costs	Overheads / Indirect Costs
Arise when one process makes many products together	Support functions needed across the organization
Only exist before split-off point	Exist everywhere in the organization
Related to production	Related to general operations

4.4.3 Allocation Methods

1. **Sales Value at Split-Off Method** (preferred when prices available at split-off):

Sales Value at Split-Off

$$\text{Allocation to Product}_i = \text{Joint Cost} \times \frac{\text{Sales Value of Product}_i \text{ at Split-Off}}{\text{Total Sales Value of All Products at Split-Off}}$$

Follows the *benefits-received* criterion. Uses total production, not just quantity sold.

2. **Physical-Measure Method:**

Physical Measure Method

$$\text{Allocation to Product}_i = \text{Joint Cost} \times \frac{\text{Physical Measure of Product}_i}{\text{Total Physical Measure of All Products}}$$

Uses weight, quantity, or volume at the split-off point.

3. **Net Realizable Value (NRV) Method:** Used when no market price exists at split-off.

$$\text{NRV}_i = \text{Final Sales Value}_i - \text{Separable Costs}_i$$

Quality Chicken (E17-21)

Joint cost = \$50. Using **Sales Value Method**:
Total sales value = $(100\times0.55)+(20\times0.20)+(40\times0.35)+(80\times0.10)+(10\times0.05) = \$81.50$
Breasts allocation: $\$50 \times \frac{55.00}{81.50} = \$33.74$, so 40 lbs destroyed = $\$33.74 \times \frac{40}{100} = \$13.50$
Wings allocation: $\$50 \times \frac{4.00}{81.50} = \$2.45$, so 15 lbs destroyed = $\$2.45 \times \frac{15}{20} = \$1.84$
Total insurance claim (Sales Value): $\$13.50 + \$1.84 = \$15.34$

If selling price at split-off is available, the sales value at split-off method is preferred because it provides the best measure of benefits received, is independent of further processing decisions, uses a common basis (revenue), and is simple to apply.

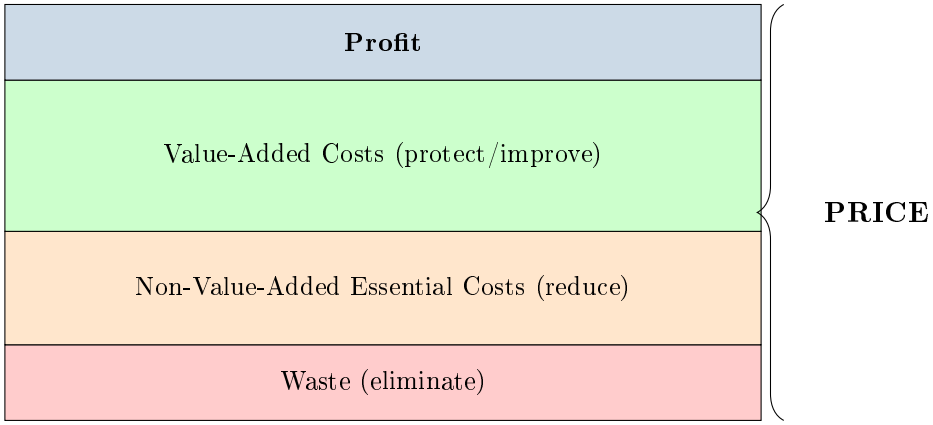
5 Capacity Analysis and Pricing (Session 6)

5.1 The Link Between Costs and Pricing

Pricing decisions are fundamental to managerial accounting. The price has a "boundary"—you cannot simply raise it whenever costs rise. Therefore, improving profit typically means working on the cost side:

- **Protect/improve** value-added efficiency and quality
- **Reduce** non-value-added essential costs where possible
- **Eliminate** waste aggressively

5.2 Price Components



5.3 Four Denominator-Level Capacity Concepts

The "denominator level" is the quantity used in the denominator when calculating the fixed overhead rate. This choice has **major strategic implications**.

Fixed Overhead Rate

$$\text{Fixed OH Rate} = \frac{\text{Budgeted Fixed Manufacturing OH Costs}}{\text{Denominator-Level Capacity}}$$

Capacity Level	Definition
Theoretical	Maximum (ideal) capacity: producing at full efficiency all the time. Unattainable in practice.
Practical	Theoretical capacity reduced by unavoidable operating interruptions (maintenance, breakdowns).
Normal	Average customer demand over *several periods*. Includes seasonal, cyclical, and trend factors.
Master-Budget	Expected capacity utilization for the *next budget period* (typically one year).

Thunder Bolt Inc. (E9-30)

Budgeted Fixed MOH = \$6,480,000.

Capacity Level	Units	Fixed OH Rate
Theoretical ($3 \times 5 \times 360$)	5,400	\$6,480,000 / 5,400 = \$1,200
Practical ($3 \times 4 \times 320$)	3,840	\$6,480,000 / 3,840 = \$1,687.50
Normal	3,240	\$6,480,000 / 3,240 = \$2,000
Master-Budget	3,600	\$6,480,000 / 3,600 = \$1,800

Notice how the unit cost **increases dramatically** as the denominator decreases.

5.4 Which Capacity Level to Use?

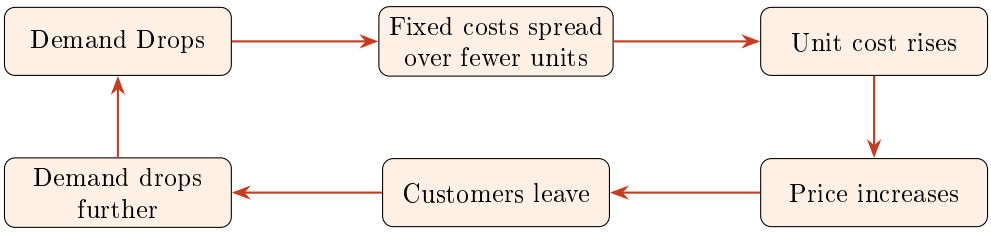
Purpose	Recommended Capacity Level
Product Costing & Capacity Management	Practical
Pricing Decisions	Practical (short-term), Normal (long-run)
Performance Evaluation	Master-Budget
External Reporting	Normal
Tax Requirements	Actual

5.5 The Downward Demand Spiral

When demand-based capacity levels (master-budget) are used for pricing:

1. Demand drops → fewer units over which to spread fixed costs
2. Unit cost increases → prices rise to cover costs
3. Higher prices → more customers leave
4. Demand drops further → costs spread over even fewer units
5. Repeat until the business becomes unviable

The Terminus Hotel (B) illustrates this: management declined offers at 90 and 95 per room because the cost per room (using low-occupancy forecast as denominator) exceeded these rates. By refusing, they lost customers, further reducing occupancy, further increasing per-unit costs—a classic death spiral.



The choice of capacity denominator has profound effects on unit costs and pricing. Using practical capacity for product costing separates the cost of *using* capacity from the cost of *having* unused capacity, preventing the death spiral. Always consider the broad impact of capacity assumptions on decision-making.

6 Cost Refinement and Activity-Based Costing (Session 7)

6.1 Why Refine the Costing System?

Simple costing systems using broad averages can produce inaccurate product costs:

- **Product Undercosting:** A product consumes a high level of resources but is reported with a low cost per unit.
- **Product Overcosting:** A product consumes a low level of resources but is reported with a high cost per unit.
- **Product-cost cross-subsidization:** When undercosting one product leads to overcosting another.

6.1.1 Signals That Cost Refinement Is Needed

1. Significant overhead costs allocated using one or two cost pools
2. Most overhead is considered unit-level
3. Products consume different amounts of resources
4. Products that should be profitable consistently show small profits
5. Operations staff disagrees with accounting over cost figures

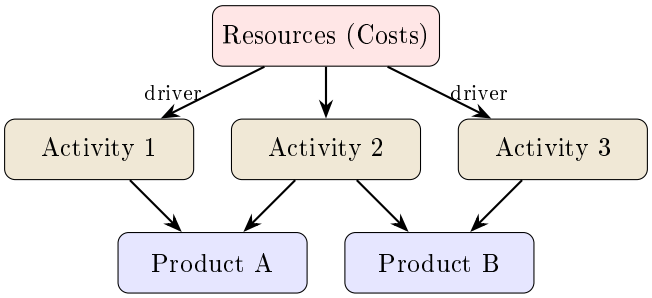
6.2 Activity-Based Costing (ABC)

ABC Foundation

Traditional approach: Products → demand costs.

ABC approach: Products → demand **activities**; Activities → demand costs.

ABC uses *multiple* cost pools organized around activities, with *activity-specific* cost drivers as allocation bases.



6.3 ABC Cost Hierarchies

Level	Description	Examples
Unit-level	Activities performed on each individual unit	Direct materials, machine energy
Batch-level	Activities associated with a group of units	Setup costs, quality inspections per batch
Product-sustaining	Activities to support a product regardless of units/batches	Product design, engineering changes
Facility-sustaining	Activities supporting the organization as a whole	Plant security, building lease, general admin

6.4 Time-Driven Activity-Based Costing (TDABC)

TDABC simplifies ABC by requiring only **two inputs**:

TDABC Inputs

1. **Cost per time unit of supplying resource capacity:**

$$\text{Cost Rate} = \frac{\text{Total Cost of Resources Supplied}}{\text{Practical Capacity of Resources (in time units)}}$$

2. **Unit times of consumption:** How much time each product, service, or customer uses.

$$\text{Cost Assigned} = \text{Cost Rate per Time Unit} \times \text{Time Required}$$

ABC is generally perceived to produce superior costing figures due to the use of multiple drivers across multiple cost hierarchy levels. However, ABC is only as good as the drivers selected and their actual relationship to costs. Poorly chosen drivers produce inaccurate costs even with ABC.

7 Budgeting: Planning, Control, and Decision-Making (Session 9)

7.1 What Is a Budget?

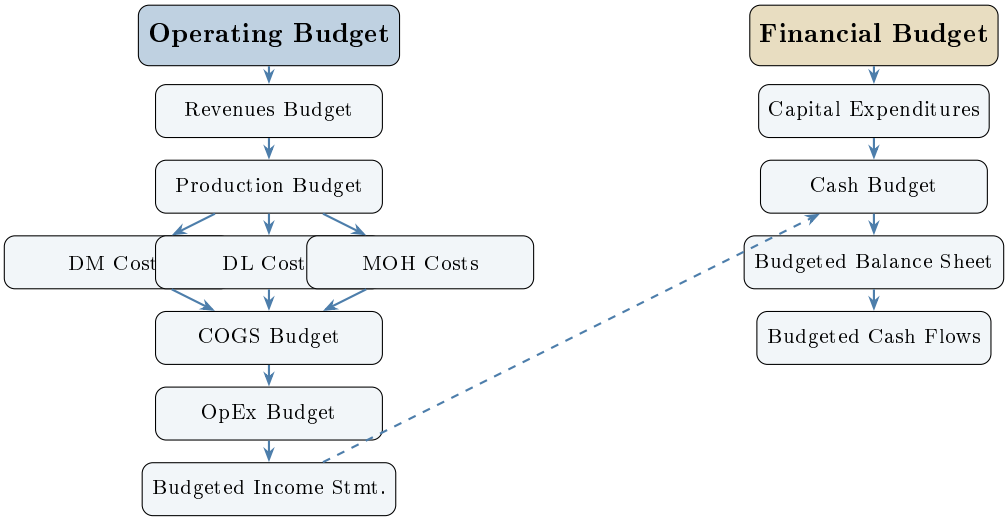
A budget is the **quantitative expression of a proposed plan of action** by management for a specified period. It serves as an aid to coordinating what needs to be done and may include both financial and nonfinancial data.

7.2 Purposes of Budgeting

1. **Planning:** Translating strategy into actionable plans.
2. **Facilitating Communication and Coordination:** Aligning departments.
3. **Allocating Resources:** Directing resources to strategic priorities.
4. **Controlling Profit and Operations:** Setting benchmarks.
5. **Evaluating Performance and Providing Incentives:** Measuring results.

7.3 The Master Budget

The master budget consists of two main components:



7.4 Key Budget Formulas

Production Budget

$$\text{Units to Produce} = \text{Budgeted Sales} + \text{Desired Ending Inventory} - \text{Beginning Inventory}$$

Direct Materials Purchases Budget

$$\text{DM Needed for Production} = \text{Units to Produce} \times \text{DM per Unit}$$
$$\text{DM to Purchase} = \text{DM Needed} + \text{Desired Ending DM Inv.} - \text{Beginning DM Inv.}$$

Manufacturing Overhead Budget

$$\text{Total MOH} = (\text{Variable MOH Rate} \times \text{DLH}) + \text{Fixed MOH}$$
$$\text{MOH Rate} = \frac{\text{Total MOH}}{\text{Total DLH}}$$

Unit Manufacturing Cost

$$\text{Unit Cost} = \text{DM per unit} + \text{DL per unit} + \text{MOH per unit}$$

7.5 Budget Preparation Techniques

Technique	Description
Budget Lapsing	Unspent amounts cannot be carried over to the subsequent year.
Zero-Based (ZBB)	Must justify every line item from zero each year. No automatic carry-forward.
Incremental	Previous year adjusted for expected changes in volume and prices.

7.6 Behavioral Considerations

- **Top-Down:** Budget imposed by senior management. Can lack buy-in from lower levels.
- **Participative (Bottom-Up):** Input from all levels. Better buy-in but risk of *budgetary slack*.
- **Pseudo-Participative:** Appears participative but senior management overrides input.
- **Budgetary Slack:** Subordinates deliberately underestimate revenues or overestimate costs to make targets easier to achieve.

7.7 Static vs. Flexible Budgets

Static Budget	Flexible Budget
Based on one planned level of activity	Adjusts for actual activity level achieved
Limited use for performance evaluation	Better for control and evaluation
Cannot isolate efficiency from volume effects	Separates price and quantity variances

Variance Analysis

Price Variance: Evaluates the cost of resources consumed.

$$\text{Price Variance} = (\text{Actual Price} - \text{Budgeted Price}) \times \text{Actual Quantity}$$

Quantity (Efficiency) Variance: Evaluates the amount of resources consumed.

$$\text{Quantity Variance} = (\text{Actual Qty} - \text{Budgeted Qty for Actual Output}) \times \text{Budgeted Price}$$

Budgets should focus activities and resources on organizational goals, not merely limit spending. The key is the quality of **assumptions**. Flexible budgets and variance analysis are superior to static budgets for performance evaluation because they separate volume effects from efficiency.

8 Inventory Costing: Variable vs. Absorption (Session 10)

8.1 Three Costing Methods

Method	Costs Included as Product (Inventoriable) Costs
Absorption (Full)	DM + DL + Variable MOH + **Fixed MOH**
Variable (Direct)	DM + DL + Variable MOH (Fixed MOH is a period cost)
Supervariable (Throughput)	DM only (all others are period costs)

8.2 Income Statement Formats

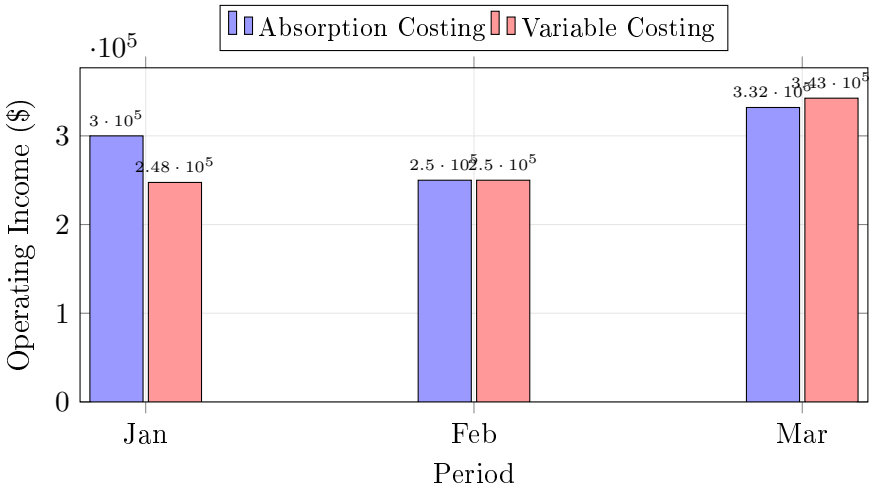
Variable Costing	Absorption Costing
Revenues − Variable COGS − Variable Period Costs = **Contribution Margin** − Fixed Product Costs (Fixed MOH) − Fixed Period Costs = **Operating Income**	Revenues − COGS (includes Fixed MOH) = **Gross Margin** − Variable Period Costs − Fixed Period Costs = **Operating Income**

8.3 Key Relationship: Production vs. Sales

Income Comparison Rule

- If **Production** > **Sales**: Absorption OI > Variable OI (fixed costs "hide" in inventory)
- If **Production** < **Sales**: Absorption OI < Variable OI (fixed costs released from inventory)
- If **Production** = **Sales**: Absorption OI = Variable OI

  Difference in OI = (Units Produced − Units Sold) × Fixed MOH per Unit



8.4 Performance Issues with Absorption Costing

Under absorption costing, managers can **manipulate income by overproducing**:

1. Production beyond demand increases inventory
2. More fixed costs are capitalized (deferred) in inventory
3. Fewer fixed costs are expensed in the current period
4. Reported profit increases—even though no additional sales occurred

This is exactly what happened in the Sunbeam case under CEO Albert Dunlap.

Quick Numerical Comparison

Given: DM = 40,000; DL = 30,000; Variable MOH = 20,000; Fixed MOH = 10,000. Units produced = 1,000; Units sold = 700; SP = 110/unit.

Absorption: Unit cost = $\frac{40{,}000+30{,}000+20{,}000+10{,}000}{1{,}000}$ =100/unit
COGS = 700 × 100 =70,000. Gross Margin = 77,000 − 70,000 =7,000.

Variable: Unit cost = $\frac{40{,}000+30{,}000+20{,}000}{1{,}000}$ =90/unit
Variable COGS = 700 × 90 =63,000. CM = 77,000 − 63,000 =14,000.
Less Fixed MOH (10,000) = OI = 4,000.

Difference: 7,000 − 4,000 = 3,000 = (1,000 − 700) × $\frac{10{,}000}{1{,}000}$ = 300×10.

Variable costing separates fixed manufacturing costs from product costs, making it better for internal decision-making. Absorption costing is required for external reporting (GAAP/IFRS). The choice of method affects reported income when production ≠ sales.

9 Cost-Volume-Profit (CVP) Analysis (Session 11)

9.1 What Is CVP Analysis?

CVP analysis examines the relationships between **revenues, costs, volume, and profits**. It is one of the most powerful tools for short-term planning and decision-making.

9.2 The CVP Equation

Core CVP Equation

$$\boxed{\text{Revenue} - \text{Variable Costs} - \text{Fixed Costs} = \text{Operating Income}}$$

Expanded:

$$(P \times Q) - (V \times Q) - FC = OI$$

Where: P = Selling price per unit, Q = Quantity of units sold, V = Variable cost per unit, FC = Total fixed costs, OI = Operating income.

Contribution Margin

$$\text{Contribution Margin per Unit (CM)} = P - V$$

$$\text{Contribution Margin Ratio (CMR)} = \frac{P - V}{P} = \frac{\text{CM}}{P}$$

$$\text{Total CM} = \text{Revenue} - \text{Total Variable Costs} = CM \times Q$$

$$\text{Operating Income} = \text{Total CM} - FC$$

9.3 CVP Assumptions

1. Total revenues behave linearly (Revenue $= P \times Q$)
2. Total expenses behave linearly (TC $= FC + V \times Q$)
3. Selling price, variable cost per unit, and fixed costs are known and constant
4. All units produced are sold in the same period
5. Single product analysis (or constant sales mix for multiple products)
6. Time value of money is ignored

9.4 Break-Even Analysis

The break-even point (BEP) is where operating income equals zero.

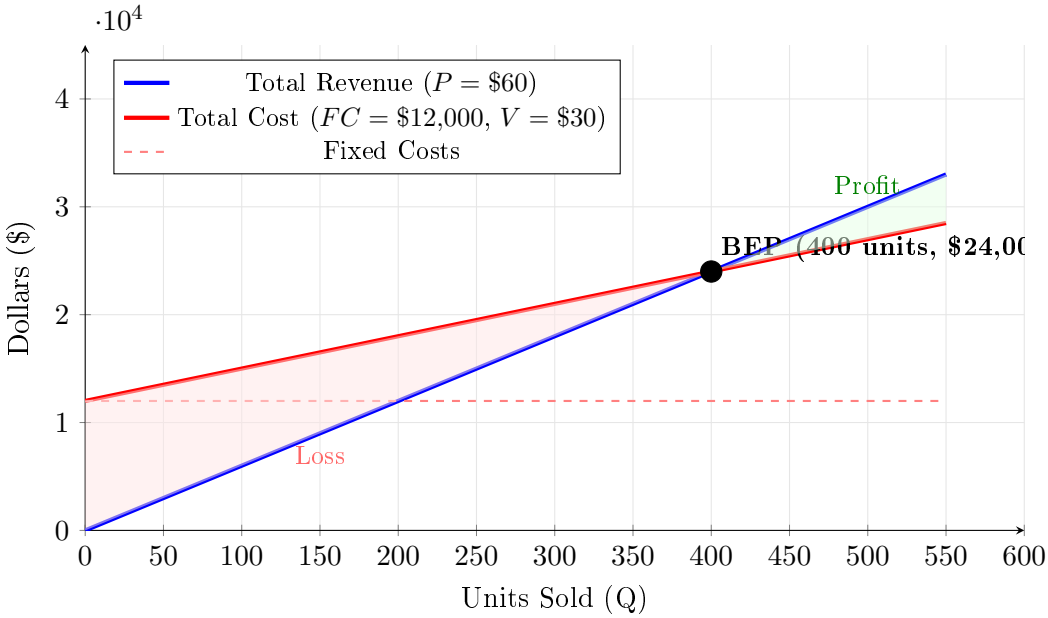
Break-Even Formulas

Break-Even in Units:

$$Q_{\text{BE}} = \frac{FC}{CM} = \frac{FC}{P - V}$$

Break-Even in Revenue:

$$\text{Revenue}_{\text{BE}} = \frac{FC}{CMR} = \frac{FC}{\frac{P-V}{P}}$$



9.5 Target Profit Analysis

Target Operating Income

Units needed for target OI:

$$Q_{\text{target}} = \frac{FC + \text{Target OI}}{CM}$$

Revenue needed for target OI:

$$\text{Revenue}_{\text{target}} = \frac{FC + \text{Target OI}}{CMR}$$

9.6 CVP and Income Taxes

After-Tax Profit

$$\text{Net Income (NI)} = OI \times (1 - \text{Tax Rate})$$

$$OI = \frac{NI}{1 - \text{Tax Rate}}$$

To find units for a **target net income**:

$$Q = \frac{FC + \frac{\text{Target NI}}{1-t}}{CM}$$

9.7 Margin of Safety

The margin of safety measures how far current or expected sales are above the break-even point.

Margin of Safety

$$\text{Margin of Safety (\$)} = \text{Actual (or Expected) Revenue} - \text{Break-Even Revenue}$$

$$\text{Margin of Safety (\%)} = \frac{\text{Margin of Safety (\$)}}{\text{Actual (or Expected) Revenue}} \times 100$$

Dogfish Head Brewery

$P = \$100/\text{case}$, $V = \$62/\text{case}$ (30+16+6+10), Revenue = \$7,000,000, OI = \$800,000.

$CM = 100 - 62 = \$38/\text{case}$; $CMR = 38/100 = 0.38$

Total CM = $7{,}000{,}000 \times 0.38 = \$2{,}660{,}000$

$FC = 2{,}660{,}000 - 800{,}000 = \$1{,}860{,}000$

$\text{Revenue}_{BE} = \frac{1{,}860{,}000}{0.38} = \$4{,}894{,}737$

Margin of Safety = $7{,}000{,}000 - 4{,}894{,}737 = \$2{,}105{,}263$

Margin of Safety % = $\frac{2{,}105{,}263}{7{,}000{,}000} = 30.1\%$

Sales can decrease 30% before the company experiences a loss.

9.8 Sensitivity Analysis and Operating Leverage

Sensitivity analysis examines "what-if" scenarios: what happens to profit if selling price, volume, or cost structure changes?

Degree of Operating Leverage (DOL)

$$DOL = \frac{\text{Contribution Margin}}{\text{Operating Income}}$$

The DOL indicates how sensitive operating income is to changes in sales:

$$\%\Delta OI = DOL \times \%\Delta\text{Sales}$$

	Bogside Farm	**Sterling Farm**
Sales	\$100,000 (100%)	\$100,000 (100%)
Variable Costs	\$60,000 (60%)	\$30,000 (30%)
Contribution Margin	\$40,000 (40%)	\$70,000 (70%)
Fixed Costs	\$30,000	\$60,000
Operating Income	\$10,000	\$10,000
DOL	40,000/10,000 = **4.0**	70,000/10,000 = **7.0**

A 10% increase in sales yields 40% increase in OI for Bogside but 70% for Sterling. **Higher fixed costs (higher DOL) means higher risk and higher reward.**

A higher proportion of fixed costs (higher DOL) amplifies both gains and losses. Companies with high operating leverage benefit more when sales increase but suffer more when sales decline. This is a critical consideration for strategic cost structure decisions.

CVP analysis provides a quantitative foundation for short-term decisions. Master the break-even, target profit, margin of safety, and operating leverage formulas. Remember that CVP relies on strict assumptions (linearity, constant mix) that must be validated against the relevant range.

10 Case Studies: Key Lessons

10.1 Ecoservicios S.L. (Session 1)

Context: A Spanish environmental consulting firm must set a price for a major contract with a public utility. Four different pricing proposals emerge from team members with different backgrounds (management, law, technical, financial).

Key Lessons:

- There is no single "scientific" cost—**different costs serve different purposes**.
- The team composition (diverse expertise) is critical for pricing decisions.
- Daniel Peralta's proposal: use subsidized rates for idle hours already committed ⇒ 246,843.
- Antonio Martínez's proposal: zero-price idle hours to compete on price ⇒ 214,173.
- Celia Martos's approach: target costing from the client's perspective, then restructure internal costs ⇒ 210,000–220,000.
- Carlos Romera's confusion illustrates that engineers (and others) often expect a single "correct" answer.

10.2 Terminus Hotel A (Sessions 4–5)

Context: Profitability analysis of three services (accommodation, restaurant, entertainment) using different cost allocation approaches.

Key Lessons:

- Single allocation bases (employees or revenue) produce different and potentially misleading profitability pictures.
- Multiple allocation bases (matching cost drivers to cost behavior) provide more accurate results.
- Allocation method choice can determine whether a service appears profitable or unprofitable.

10.3 Terminus Hotel B (Session 6)

Context: Hotel with 200 rooms and 2,500,000 in fixed costs faces low occupancy forecasts. Declined bids at 90 and 95 per room.

Key Lessons:

- The hotel used demand-based capacity (40% occupancy = 80 rooms/day) as the denominator.
- Unit cost: $\frac{2{,}500{,}000}{80\times365} + 25 \approx \$85.62 + \$25 = \$110.62$/room/day.
- Both bids (90 and 95) exceeded variable cost (25) and contributed to fixed cost coverage.
- By declining, the hotel **entered a downward demand spiral**.
- Lesson: Use practical capacity for pricing; accept any price above variable cost when there is excess capacity.

10.4 Enred@me (Session 7)

Context: Internet service provider analyzing subscriber profitability using simple vs. ABC costing.

Key Lessons:

- Simple costing (single allocation base: number of subscribers) masks true profitability differences between customer types.
- ABC reveals that different customer types (big associations, small associations, individuals) consume resources very differently.
- Pricing and strategic decisions should be based on refined (ABC) cost information.

11 Comprehensive Formula Reference

Cost Behavior

$$\begin{aligned}
\text{Total Variable Cost} &= V \times Q \\
\text{Total Fixed Cost} &= FC \quad \text{(constant within relevant range)} \\
\text{Total Cost} &= FC + V \times Q \\
\text{Mixed Cost} &= a + bX \quad \text{(semi-variable)}
\end{aligned}$$

Cost of Goods Manufactured & Sold

$$\begin{aligned}
\text{DM Used} &= \text{Beg. DM Inv.} + \text{DM Purchases} - \text{End. DM Inv.} \\
\text{Total Mfg. Cost} &= \text{DM Used} + \text{DL} + \text{MOH} \\
\text{COGM} &= \text{Beg. WIP} + \text{Total Mfg. Cost} - \text{End. WIP} \\
\text{COGS} &= \text{Beg. FG} + \text{COGM} - \text{End. FG}
\end{aligned}$$

Overhead Rates

$$\begin{aligned}
\text{Budgeted OH Rate} &= \frac{\text{Budgeted MOH}}{\text{Budgeted Allocation Base}} \\
\text{OH Allocated} &= \text{Budgeted Rate} \times \text{Actual Allocation Base} \\
\text{Under/Over-Allocated} &= \text{Actual MOH} - \text{OH Allocated}
\end{aligned}$$

Capacity-Based Fixed OH Rate

$$\text{Fixed OH Rate} = \frac{\text{Budgeted Fixed MOH}}{\text{Denominator Capacity Level}}$$

Joint Cost Allocation

$$\begin{aligned}
\text{Sales Value Method:} \quad \text{Alloc}_i &= \text{Joint Cost} \times \frac{\text{Sales Value}_i}{\sum \text{Sales Values}} \\
\text{Physical Measure:} \quad \text{Alloc}_i &= \text{Joint Cost} \times \frac{\text{Physical Qty}_i}{\sum \text{Physical Qtys}} \\
\text{NRV}_i &= \text{Final Sales Value}_i - \text{Separable Costs}_i
\end{aligned}$$

Budgeting

$$\begin{aligned}
\text{Production (units)} &= \text{Budgeted Sales} + \text{End. FG Inv.} - \text{Beg. FG Inv.} \\
\text{DM Purchases} &= \text{DM for Production} + \text{End. DM Inv.} - \text{Beg. DM Inv.} \\
\text{Price Variance} &= (AP - BP) \times AQ \\
\text{Quantity Variance} &= (AQ - BQ_{\text{allowed}}) \times BP
\end{aligned}$$

Inventory Costing Income Reconciliation

$$\text{Absorption OI} - \text{Variable OI} = (Q_{\text{produced}} - Q_{\text{sold}}) \times \text{Fixed MOH per unit}$$

CVP Analysis

$$\begin{aligned}
CM &= P - V \quad ; \quad CMR = \frac{CM}{P} \\
Q_{BE} &= \frac{FC}{CM} \quad ; \quad \text{Rev}_{BE} = \frac{FC}{CMR} \\
Q_{\text{target}} &= \frac{FC + \text{Target OI}}{CM} \\
Q_{\text{after-tax}} &= \frac{FC + \frac{\text{Target NI}}{1-t}}{CM} \\
\text{MoS (\$)} &= \text{Revenue} - \text{Rev}_{BE} \\
\text{MoS (\%)} &= \frac{\text{MoS (\$)}}{\text{Revenue}} \\
DOL &= \frac{\text{Total CM}}{OI} \\
\%\Delta OI &= DOL \times \%\Delta\text{Revenue}
\end{aligned}$$

12 Practice Problems with Worked Solutions

12.1 Problem 1: Cost Classification

Cooper Furniture makes recliners and chairs. Classify each cost as Direct (D) or Indirect (I), and Variable (V) or Fixed (F), with respect to the recliner product line.

Solution

Cost Item	D/I	V/F
A. Fabric used on recliners	D	V
B. Salary of PR manager	I	F
C. Annual convention attendance	I	F
D. Lubricant for assembly line	I	V
E. Freight costs for recliner frames	D	V
F. Electricity for assembly line (entire plant)	I	V
G. Wages of temporary assembly workers	D	V
H. Annual fire insurance for plant	I	F
I. Wages of plant manager (both lines)	I	F

12.2 Problem 2: Normal vs. Actual Costing

Given: Budgeted MOH = \$5,100,000; Budgeted DLH = 150,000. Actual MOH = \$5,355,000; Actual DLH = 153,000.

Solution

(a) Budgeted Rate: $\frac{\$5{,}100{,}000}{150{,}000} = \$34/\text{DLH}$

(b) Actual Rate: $\frac{\$5{,}355{,}000}{153{,}000} = \$35/\text{DLH}$

They differ because actual overhead spending was higher than budgeted, and actual hours differed from budgeted.

Overhead allocated: $153{,}000 \times \$34 = \$5{,}202{,}000$

Under-allocated: $\$5{,}355{,}000 - \$5{,}202{,}000 = \$153{,}000$ (unfavorable)

12.3 Problem 3: Break-Even and Target Profit

A company sells products at \$80/unit with variable costs of \$50/unit and fixed costs of \$180,000.

Solution

$CM = 80 - 50 = \$30$; $CMR = 30/80 = 37.5\%$

Break-Even (units): $Q_{BE} = \frac{180{,}000}{30} = 6{,}000$ units

Break-Even (revenue): $\text{Rev}_{BE} = \frac{180{,}000}{0.375} = \$480{,}000$

Target OI = \$60,000: $Q = \frac{180{,}000+60{,}000}{30} = 8{,}000$ units

Target NI = \$45,000 (tax rate 25%): $OI = \frac{45{,}000}{1-0.25} = \$60{,}000$; $Q = \frac{180{,}000+60{,}000}{30} =$ 8,000 units

12.4 Problem 4: Variable vs. Absorption Costing

A company produces 1,000 units, sells 800. DM = \$20/unit, DL = \$15/unit, VMOH = \$10/unit, Fixed MOH = \$50,000 total. SP = \$90/unit. Variable selling = \$5/unit sold. Fixed selling = \$8,000.

Solution

Absorption Costing:
Unit product cost $= 20 + 15 + 10 + \frac{50{,}000}{1{,}000} = \$95$
Revenue $= 800 \times 90 = \$72{,}000$
COGS $= 800 \times 95 = \$76{,}000$
Gross Margin $= 72{,}000 - 76{,}000 = -\$4{,}000$
Operating Expenses $= (800 \times 5) + 8{,}000 = \$12{,}000$
Operating Income $= -4{,}000 - 12{,}000 = -\$16{,}000$

Variable Costing:
Unit product cost $= 20 + 15 + 10 = \$45$
Revenue $= \$72{,}000$
Variable COGS $= 800 \times 45 = \$36{,}000$
Variable Selling $= 800 \times 5 = \$4{,}000$
CM $= 72{,}000 - 36{,}000 - 4{,}000 = \$32{,}000$
Fixed Costs $= 50{,}000 + 8{,}000 = \$58{,}000$
Operating Income $= 32{,}000 - 58{,}000 = -\$26{,}000$

Reconciliation: $-16{,}000 - (-26{,}000) = \$10{,}000 = (1{,}000 - 800) \times 50 = 200 \times \$50$ ✓

12.5 Problem 5: Capacity Denominator Effects

Fixed MOH = \$2,000,000. Calculate the fixed OH rate under:

Solution

Capacity Level	Units	Rate per Unit
Theoretical	10,000	\$200.00
Practical	8,000	\$250.00
Normal	6,000	\$333.33
Master-Budget	5,000	\$400.00

Observation: The lower the denominator, the higher the unit cost. Using master-budget capacity in a low-demand period makes products appear very expensive, potentially triggering a downward demand spiral if used for pricing.

End of Study Guide — Good luck on your exam!